

INTERMEDIATE PROGRAMMER · FINAL PROJECT

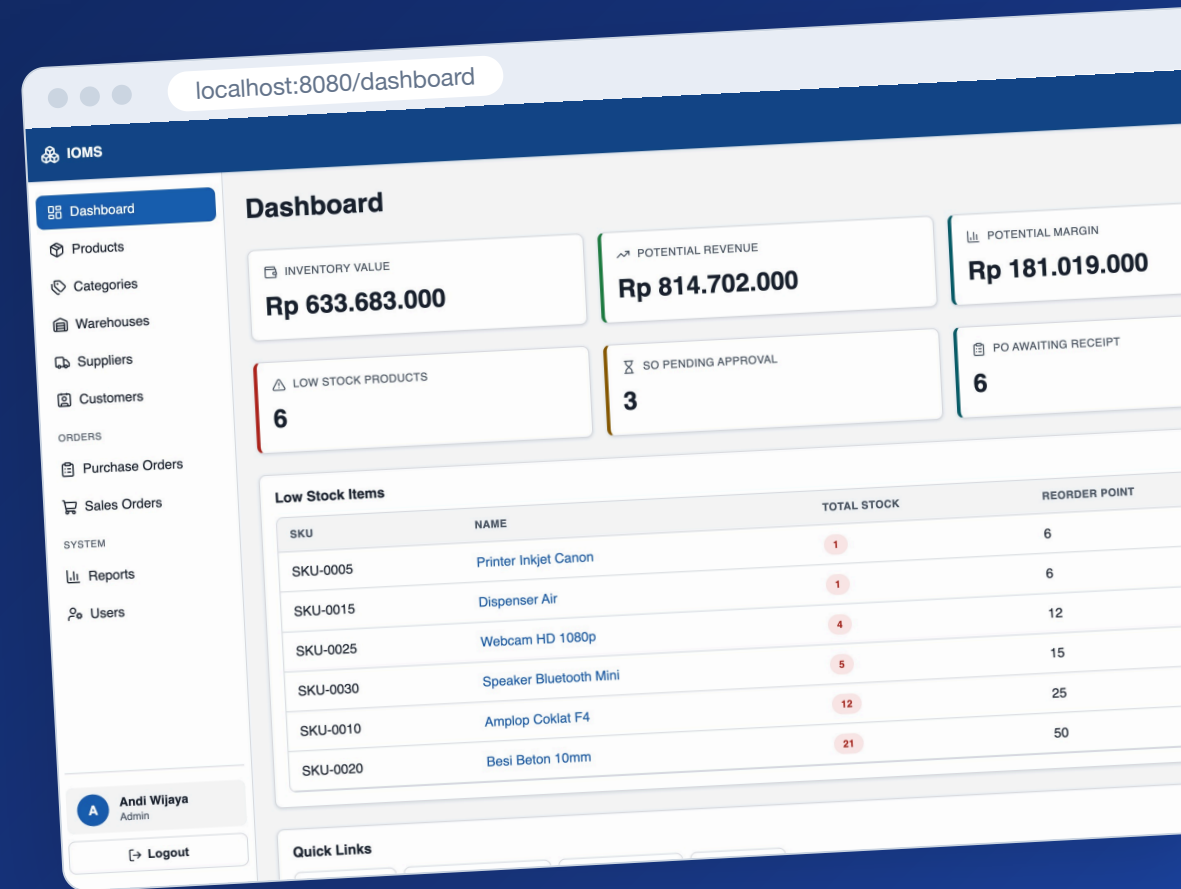
Inventory & Order Management System

Stok multi-gudang yang konsisten, termasuk saat ada proses yang berjalan bersamaan.

DISUSUN OLEH
Arief Aryudi Syidik

PROGRAM
PT Neuronworks Indonesia

SESI
Rabu, 7 Oktober 2026



Satu sistem, tiga peran, stok tercatat konsisten

Tim gudang dan sales mencatat pembelian, penjualan, dan stok di beberapa gudang. Angka stoknya perlu tetap konsisten, termasuk saat dua proses berjalan bersamaan.



Admin

Mengelola master data dan user. Satu-satunya yang boleh menyetujui SO, tapi tidak untuk SO buatannya sendiri.



Sales

Membuat dan mengajukan SO, hanya bisa melihat SO miliknya sendiri, dan bisa cek stok langsung.



Warehouse Staff

Memproses barang masuk (PO) dan barang keluar (SO); stok dan ledger ikut terupdate dalam satu transaksi.

✓ **Fitur wajib tersedia**

PO + goods receipt

SO + approval + goods issue

Stock ledger

Laporan CSV

JSON endpoint

Dashboard per role

Teknologi yang digunakan



PHP 8.2 native

OOP berlapis, tanpa framework dan ORM; PDO prepared statement.



MySQL 8

Relasi, constraint, index, transaksi eksplisit.



Vanilla JS + Fetch

HTML semantik, CSS buatan sendiri, ikon Lucide (SVG lokal).



Docker Compose

App (Apache non-root) + MySQL; schema dan seed jalan otomatis.

Purchase Order → penerimaan barang

Draft > Ordered > Partially > Received

- 1 Warehouse membuat PO (SKU-0001, qty 10) lalu klik **Order**.
- 2 **Receive** sebagian (4) → status PartiallyReceived.
- 3 Stok produk naik 4 dan muncul baris `stock_ledger` tipe Receipt, semuanya dalam **satu transaksi**.

Perubahan data ditampilkan langsung dari database.

The screenshot shows the IOMS web application interface for Purchase Orders. The browser address bar displays 'localhost:8080/purchase-orders'. The left sidebar contains navigation links for Dashboard, Products, Categories, Warehouses, Suppliers, Customers, and a section for ORDERS (Purchase Orders, Sales Orders) and SYSTEM (Reports, Users). The main content area is titled 'Purchase Orders' and includes a '+ New Purchase Order' button. Below this is a search and filter section with fields for Search (Order number or name), Status (All), Supplier (All), and Sort by Date (Newest first), along with an 'Apply' button. The main table lists 12 purchase orders with columns for PO NUMBER, SUPPLIER, ORDER DATE, and STATUS. The status values are color-coded: RECEIVED (green), PARTIALLYRECEIVED (orange), ORDERED (blue), DRAFT (grey), and CANCELLED (red). Each row has a 'View' button. At the bottom, there is a pagination control showing '« Prev 1 2 Next »'.

PO NUMBER	SUPPLIER	ORDER DATE	STATUS	View
PO-2026-0014	CV Jaya Abadi	2026-10-06	RECEIVED	View
PO-2026-0013	PT Global Elektronik	2026-08-14	RECEIVED	View
PO-2026-0012	CV Jaya Abadi	2026-08-13	PARTIALLYRECEIVED	View
PO-2026-0011	PT Sumber Makmur	2026-08-12	ORDERED	View
PO-2026-0010	PT Mitra Sejahtera	2026-08-11	DRAFT	View
PO-2026-0009	UD Bangun Sentosa	2026-08-10	CANCELLED	View
PO-2026-0008	PT Global Elektronik	2026-08-09	RECEIVED	View
PO-2026-0007	CV Jaya Abadi	2026-08-08	PARTIALLYRECEIVED	View
PO-2026-0006	PT Sumber Makmur	2026-08-07	ORDERED	View
PO-2026-0005	PT Mitra Sejahtera	2026-08-06	DRAFT	View

Sales Order, approval, dan validasi stok

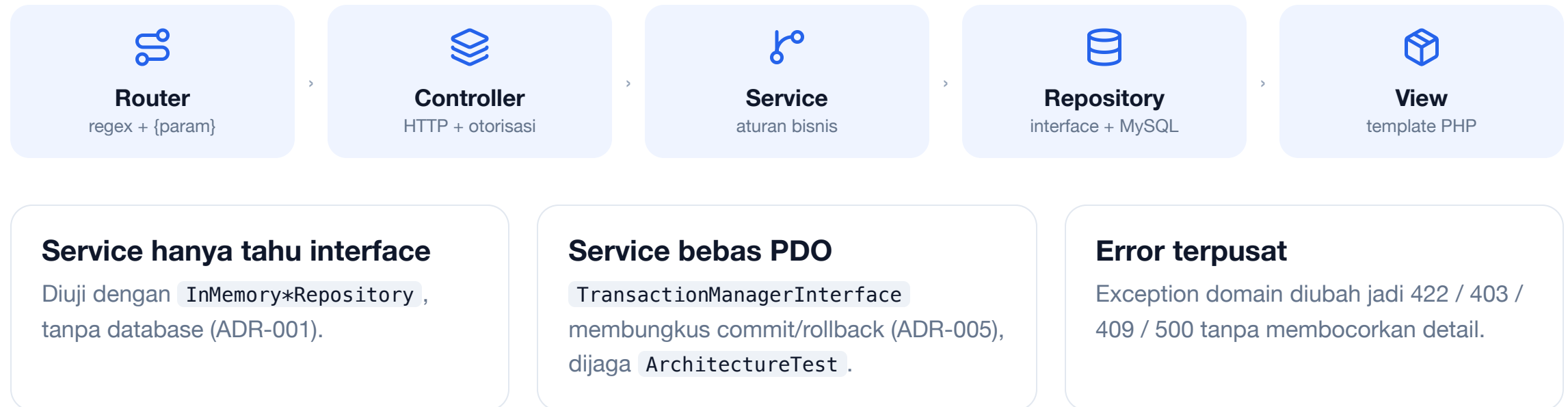
- 1 Sales membuat SO; stok per gudang terisi lewat **Fetch API** tanpa reload.
- 2 Sales mencoba approve → server membalas **403**. Admin approve.
- 3 Warehouse **Fulfill** dengan qty melebihi stok → ditolak.
- 4 Stok dan ledger **tidak berubah** (rollback).



Pemisahan tugas dicek di sisi server.

The screenshot shows a web application interface for creating a new sales order. The browser address bar displays 'localhost:8080/sales-orders/create'. The application has a dark blue header with the 'IOMS' logo. A left sidebar contains navigation links: Dashboard, Products, Customers, ORDERS (with 'Sales Orders' highlighted), SYSTEM, and Reports. The main content area is titled 'New Sales Order' and includes form fields for Customer (PT Bintang Terang), Warehouse (Gudang Pusat Jakarta), and Order Date (06/10/2026). Below these is an 'Items' table with columns for Product and Qty. One item is listed: 'Laptop Asus X441 (SKU-0001)' with a quantity of 2. A 'Remove' button is next to the item. A green box below the item shows stock information: 'Laptop Asus X441 - total stock: 56', 'In selected warehouse: 47 unit(s)', and a breakdown by warehouse: 'Gudang Cabang Surabaya: 9' and 'Gudang Pusat Jakarta: 47'. At the bottom of the form are 'Add Item', 'Save Sales Order', and 'Cancel' buttons. The footer shows the user 'Sari Dewi Sales' and a 'Logout' button.

Arsitektur berlapis dengan Dependency Inversion



Menjaga konsistensi stok

```
// SalesOrderService::fulfill()
$this->transactions->run(function () {
    foreach ($items as $item) {
        $avail = $stocks->lockForUpdate(...);
        // SELECT quantity ... FOR UPDATE
        if ($avail < $item->qty)
            throw new InsufficientStockException;
    }
    // decrement + stock_ledger + status
}); // sukses: commit · gagal: rollBack
```

- 1 **Row lock** InnoDB: dua fulfill untuk produk dan gudang yang sama harus antre.
- 2 Stok dicek **setelah** lock, jadi hasilnya pasti yang terbaru.
- 3 Satu item kurang → **seluruh transaksi dibatalkan**.
- 4 Pendekatan pessimistic lock dipilih agar pengecekan dan pengurangan stok konsisten (ADR-002).

Keamanan: kontrol yang diterapkan



CSRF

Token per sesi di semua form POST



Sesi & role

HttpOnly, SameSite, user divalidasi ulang tiap request



Brute force

Login dibatasi: 5 kali gagal per akun+IP



IDOR

Sales hanya bisa melihat SO miliknya



Injeksi

Prepared statement, output di-escape + CSP

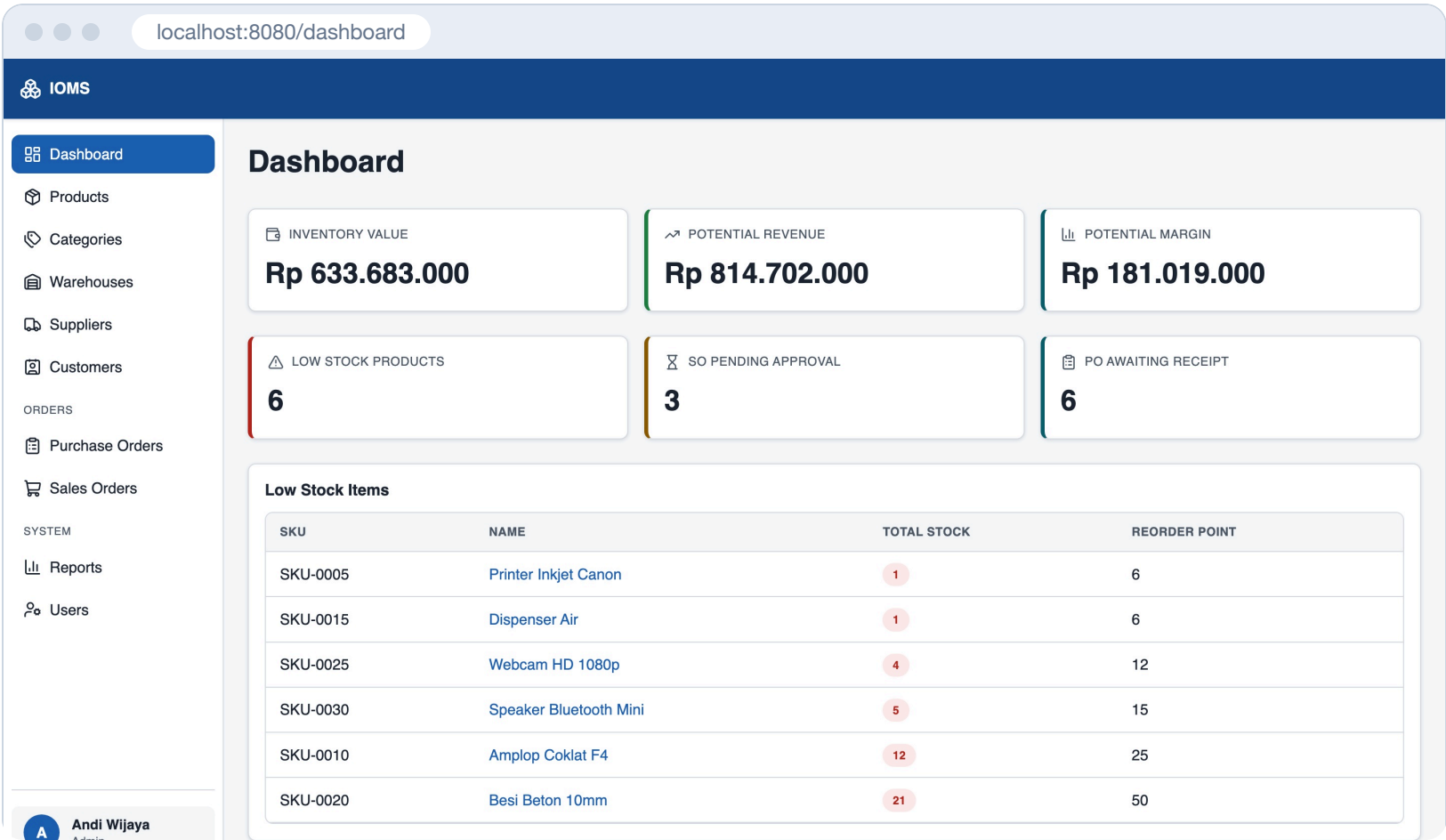


Container

Non-root, MySQL hanya di 127.0.0.1

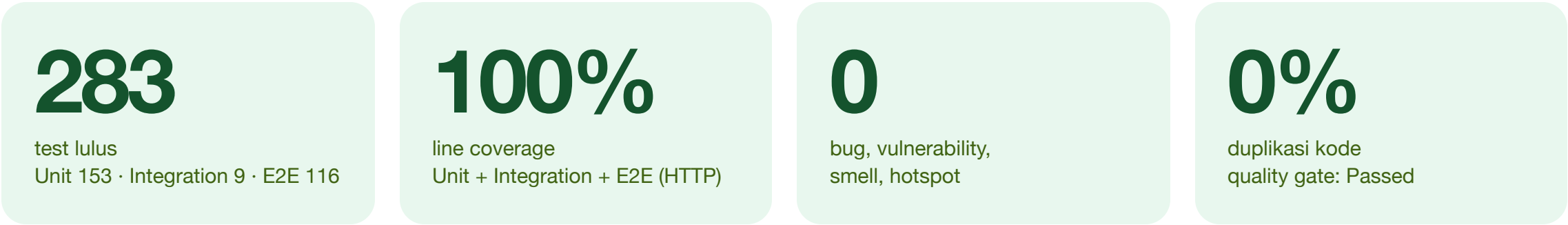
Pengembangan berikutnya: rate limiting API dan MFA. Rincian: [docs/quality/security-review.md](#).

Dashboard per role dan antarmuka konsisten



- 1 Kartu ringkasan berbeda untuk Admin, Sales, dan Warehouse.
- 2 Daftar **Low Stock** dihitung dari stok total vs reorder point.
- 3 Sidebar memuat info user dan Logout; ikon Lucide inline (SVG lokal), tanpa library JS atau CDN.

Test dan SonarQube



METRIK	SCAN AWAL	AKHIR
Isu terbuka	192	0
Coverage	21,1%	100%
Duplikasi	8,1%	0%

Coverage yang diukur adalah *line coverage*. PHPStan level 5 dan PHPCS PSR-12: 0 error.

Kendala, solusi, dan rencana pengembangan



Kendala & solusi

- **Waktu tidak sinkron:** database dan aplikasi memakai UTC, selisih 7 jam dari jam lokal.
Solusi: zona waktu PHP dan MySQL diatur ke WIB (GMT+7).
- **Library PHP hilang** saat aplikasi dijalankan dari clone baru.
Solusi: kode dan library diambil dari image Docker.



Pengembangan lanjutan

- Rate limiting dan token untuk API
- CI/CD



Prioritas

1. Rate limiting API
2. MFA dan token API

Terima kasih

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